

Humor Detection In Kannada For Sentiment Analysis

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Abstract—Sentiment analysis is one of the key trends in Natural Language processing. Sentiment analysis is extensively used by businesses in order to monitor user activity and public opinion. It is also used to analyse customer feedbacks. Humor detection is the task of identifying comical elements in textual data. It poses several challenges as it depends on the perceptions of the joke and also the context in which the joke appears. Low resource languages like Kannada pose several challenges for tasks like humor detection. These challenges include lack of labeled data, lack of humor related resources, variations in dialect and limited research. The proposed work includes manually building the humor dataset for Kannada and exploring different Machine learning and deep learning models to perform humor detection in Kannada. Of the implemented models, pre-trained Multilingual BERT(mBERT) gave the highest accuracy of 93.75 percent.

Index Terms—Natural Language processing, Data Pre-processing, Dataset, MultinomialNB, SVM, RNN-LSTM and Multilingual BERT.

I. INTRODUCTION

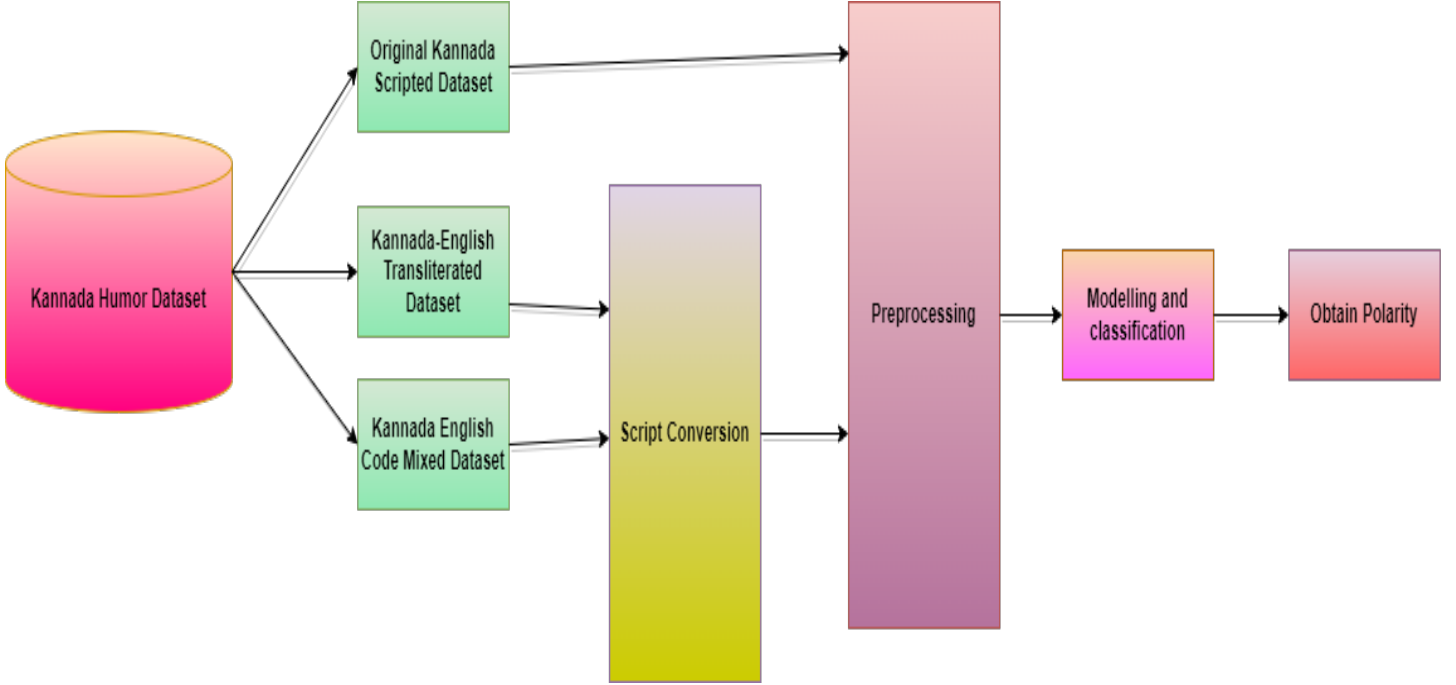
Currently huge amounts of data are available online. The problem is not about the unavailability of data, but the availability of knowledge from that data. Thus it is important to know what the data signifies. Manually classifying data would demand lot of time and effort. Thus came the necessity to automatically classify data. Sentiment analysis is a trending topic as it finds application in various places like Social media monitoring, analysis of a new product, brand monitoring and reputation management and many more. Real-time sentiment analysis of text feeds from platforms such as Twitter is immensely helpful in prompt addressing of negative or wrongful social mentions as well as threat detection in cyberbullying. In our work, we are opting to detect humor in Kannada - a morphologically rich language. Kannada is a low resource language. Low resource languages are called so because not much work is done on these languages and they lack the training data that is required to build accurate state-of-the-art NLP systems. Humor detection is the task of identifying comical or amusing elements in a text. Humor detection poses interesting linguistic challenges in NLP. It is highly subjective and depends on the perceptions of a joke and the context in which the joke appears. Further, humor detection in a low resource language like Kannada poses a bigger challenge. We have made use of both ML and DL models and evaluated their performances using metrics like Accuracy, F1-score, Precision and Recall and visualized the results in the form of heatmaps.

II. THEORY

Sentiment analysis, also referred to as opinion mining is a part of NLP that identifies the emotional tone behind the body of texts. Most of the research in field of sentiment analysis is done for wide-spoken and popular languages. Humor detection in Kannada comes with a lot of challenges like collection of dataset and pre-processing it. Due to the lack of availability of training data we created Kannada dataset from various sources. Our pre-processing tasks involve removing special characters from extracted text followed by identifying English words and translating them into Kannada. Since manually classifying the texts can be a tedious task, text classification is done involving Machine Learning and Deep learning approaches. Both ML and DL models have their own pros and cons. While DL is best suited for highly complex decisions, it also requires huge amount of dataset. On the contrary ML can be used for relatively simpler tasks and can even work on a small dataset.

III. RELATED WORK

Humor detection in other languages are found in references as mentioned below. [1] explores novel methodologies to generate and detect humor in code-mixed Hindi-English. They create embeddings by scraping a dataset of 412k Hinglish Tweets from Twitter to avoid unintelligible words and jokes due to lack of understanding of Hindi language. They have used LSTM, Bi-directional LSTM and Attention Based Bi-Directional LSTM for generating humor. LSTM, Bi-directional LSTM, IndicBERT and mBERT are used for detecting humor in code mixed Hindi-English. Of the experimented approaches, Attention Based Bi-Directional LSTM with converting parts of text on a word2vec embedding gives the best results by generating 74.8 percent good jokes and IndicBERT used for detecting humor in code - mixed Hindi-English outperforms other humor detection methods with an accuracy of 96.98 percent. [2] discusses humor detection in three languages - Chinese, Russian and Spanish. They have implemented Sentence Discrepancy Prediction i.e, paragraphs labeled as humorous(positive samples) are considered to have a setup and a punchline with large semantic discrepancy and paragraphs labeled as non-humorous(negative samples) are considered to have no setup and punchline and hence no discrepancy. They have also implemented the paragraph decomposition algorithm where a separator token is inserted in the middle/before the last sentence of the



paragraph without considering real punctuations of the paragraph. This is done in order to determine the semantic discrepancies in punchlines to detect humor.

[3] perform Sentiment Analysis in original,transliterated script and combination of both along with English.Web scraping is the main source of data presented in this work and have developed MuLSA architecture in combination with with two deep learning and one machine learning models to accomplish the required task. [4] perform Sentiment Analysis on Major Indian Languages like Kannada, Malayalam, Bengali, Konkani, Punjabi, Gujarati and Hindi. They have used SVM and Naive Bayes ans the considerable model for the dataset which they have obtained from various sources and web scraping movie reviews and YouTube for low resource language like Konkani and Punjabi.Good accuracy's are obtained for certain languages but for not all. [5] performs domain based sentiment analysis in Regional languages. They have opted popular newspapers and articles as their source of data and used only Decision Tress for modelling. [6]performs humor detection in code-mixed Hindi English tweets written in Latin scripts using models like SVM, RF and Naive Bayes Classifier. [7]performs conversational humor detection in a conversational Telugu dataset using pre-trained deep learning models such as BERT, using FastText word embeddings and Text GCN model .

IV. DATASET

Dataset used for this task is self developed.We have developed Kannada-Humor dataset which contains samples with taking Kannada language as a baseline script.It contains

samples which not only depict Kannada humor or jokes but also contain Kannada-English code mixed data.This makes our dataset more flexible and not specific to only Kannada. Dataset contains two columns.First columns is annotated as "JOKES" which contain all variety of humorous sentences and the second column is annotate as "LABEL" which has Boolean polarity. '0'and '1' correspond to Non-Humorous and Humorous respectively.Various data source are considered to develop this dataset. Scarping Twitter comments using Twitter API and hashtag names play a prominent role in web scraping comments.Data is also collected by considering images which has humorous text in it and then extracting only text from those images and feeding it to the dataset. Applications on web and Android system are used to extract Kannada jokes for developing a weighted dataset.Movie review scraping from is done for Kannada comet movies and extracted the views of viewers.Other sources of data are considered from Kannada Joke Books.

JOKES	LABEL
ಓದುವ ಆಸೆಯಿಂದ ಬೇಗಗೆ ಬೇಗ ಎದ್ದು ಅದರ ಆಸೆಯೇ ಮುಖಕ್ಕೆ ಮೂಲ ಎಂದು ವ್ಯವಾಶಾಸಿಗರ ಬಿದ್ದು	1
ಹಸಿವು ಹಸ್ಯಾಗಿದ್ದರೆ ಊಟ ಹಸ್ಯಾಗಿಯೇ ಆರುತ್ತದೆ..	0
ಶಿಕ್ಷಕರು ಹೇಳಿದ್ದು ಅತಿ ದೂಡ್ಡು ಸುಳ್ಳು 10th ಒಂದು ಪಾಠಶಾಲೆಗೆ ಸಾಕು ಮುಂದಿನ ಲೈಫ್‌ಲಾ ರಾಮಾಗಿಯೋದು	1
BATHROOM ಗೆ ಮೊದಲೇ ಬಿಟ್ಟು ಹೋದವು...?ಯುದ್ಧಕ್ಕೆ ಏಡ್ಡು ಮರೆತು ಹೋದರಾಗಿ ಫೀಲ್ ಆಗುತ್ತೆ	1

V. APPROACH

Majority of the approaches and research in Natural Language Processing techniques revolves around high-resource languages like English, Mandarin, Arabic and French, that are rich in available data. However code-mixed and code-switched narratives are customary in multi-lingual cultures, i.e. it is a natural phenomenon for individuals to code-switch words according to context, in speech and in social media sites like Facebook and Twitter. Natural Language Processing techniques for low resource languages require linguistic experts or extensive data, which is time and resource consuming to generate manually. This contributes to significant challenges in text processing.

The first step is collecting and procuring a comprehensive, balanced and appropriate dataset for the task. While some resources do exist pertaining to the low-resource language in question, they are not conducive to our needs. Therefore we conducted a data collection process to curate a dataset with accurately labeled examples for humor detection in Kannada. Next we worked on preprocessing the data and cleaning the dataset by removing noise and special characters from the text. The sentences were annotated accordingly as 'humorous' or 'non-humorous'. A key course of action while handling code-mixed text is - language identification. We defined functions to detect the language in the input code-mixed text and translated the non-Kannada words to Kannada.

We have implemented several models to detect humor in text. The models range from simple to complex and cover a variety of techniques and methodologies like Machine Learning (SVM, MNB), Deep Learning (RNN) and Transfer learning (pre-trained BERT). We used a pre-trained model for Transfer learning to leverage the knowledge learned from a vast corpus which is instrumental when there is less training data. The weights are updated to fine-tune this model utilising task-specific data.

VI. DATA PRE-PROCESSING

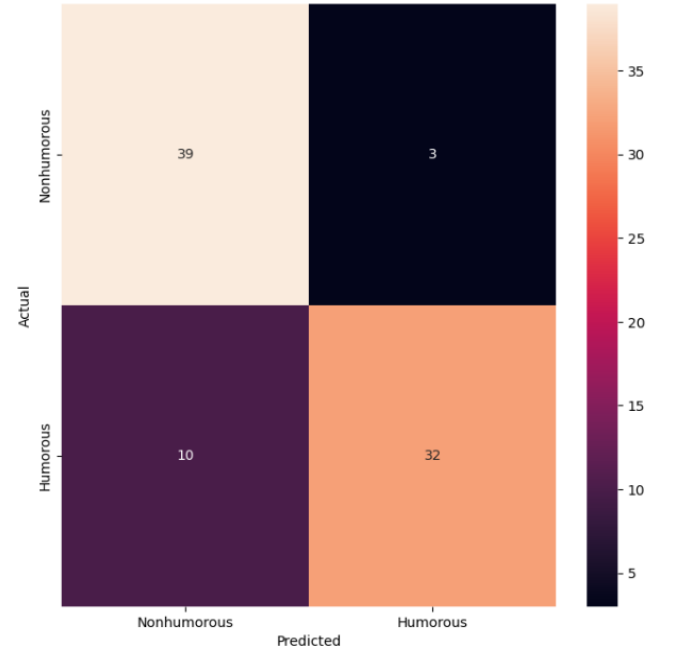
Data Preprocessing is done in two phases. The first phase includes removal of unwanted special characters. A list of special characters possible is predefined and each sample is checked for the presence of special character. Second phase of preprocessing includes conversion of data samples into base script. Kannada is considered as our baseline script. As our dataset has Kannada-English code mixed data in it, there is a need of script conversation of English to baseline script Kannada. This is accomplished in the second phase of preprocessing. Kannada language also has some stopwords in it. Stopword removal is also performed by comparing each string of the sample in dataset with the list of predefined stopwords to avoid noise in data.

VII. MODELS USED

A. MultinomialNB

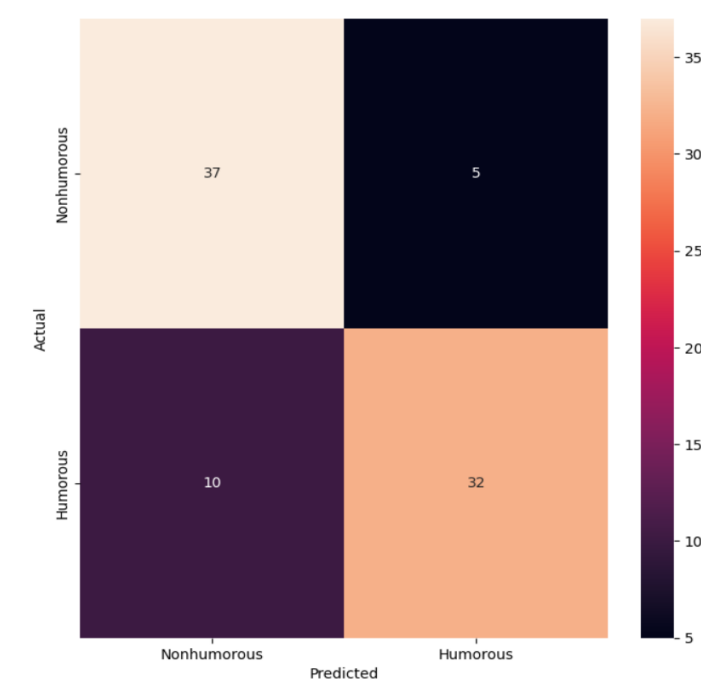
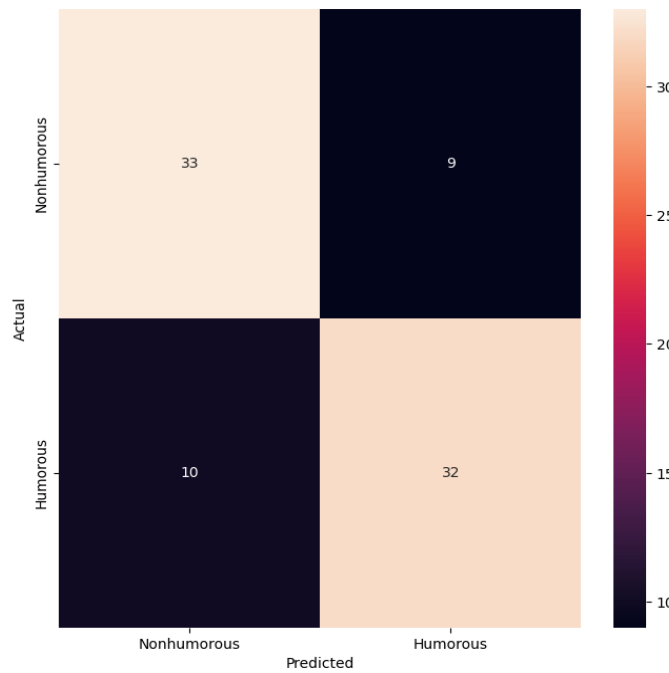
- Multinomial Naive Bayes is one of the most popular supervised learning classifications that is used for the analysis of the categorical text data.

- Dataset is split in the ratio of 80 percent for training and 20 percent for testing.
- Samples in dataset are considered and tokenized. Vectors are generated for each obtained tokens.
- The frequency of each word is computed and the results are transformed into sparse matrix representation.
- Normalization is performed on obtained matrix by converting it into Tf-IDF representation thus help in reducing the impact of frequency.
- Performing fit on the model and prediction on the test split is computed.
- Humor classification using this model lead to good result and obtained accuracy of 85 percent.



B. SVM

- Support Vector Machine is a popular Machine Learning model used for Classification and Regression. However it is primarily used for Classification problems.
- Firstly the extracted Kannada text was pre-processed by removing special characters and translating any English words into Kannada.
- The dataset is split into train and test, taking 80 percent for train and 20 percent for test.
- Next the sentences were converted into feature vectors using CountVectorizer.
- Further for constructing SVM model we used the Radial Basis Function (rbf) kernel.
- When we evaluated the model against the test data, the accuracy amounted to 77.38 percent. The results of the predictions can be visualized in the form of a heatmap.

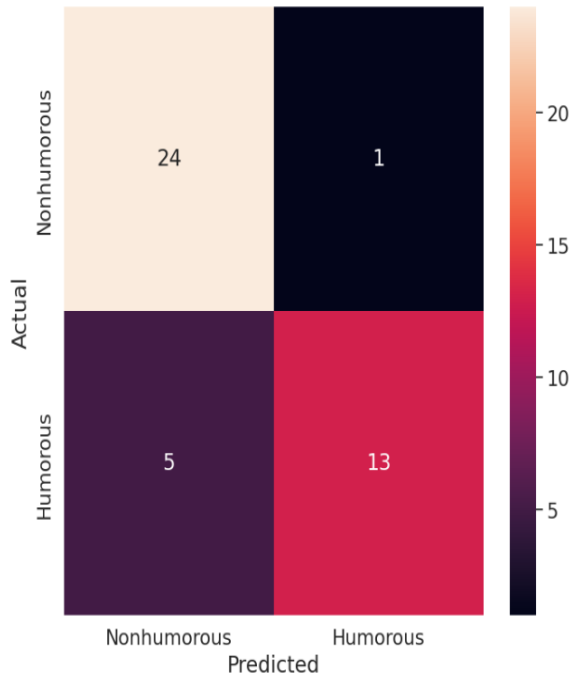


C. RNN-LSTM

- RNN-LSTM is a type of neural network that is specially used for processing textual data. It is capable of handling variable length inputs and remembers the previous input using their internal memory.
- Since humor detection takes into consideration the context in which the joke appears, the full sequence of text has to be considered.
- RNN-LSTM model can handle the complexity of the language by helping us to capture the unique syntax and grammatical rules of the language in order to make more accurate predictions.
- We split the dataset for training and testing in the ratio of 80 to 20.
- To build the RNN-LSTM model we initialize an empty sequential model. To this sequential model, we add an embedding layer and a Spatial dropout layer. This layer drops out entire 1D feature maps of the input. We also add an LSTM layer to model the sequential data.
- Finally, we add a dense layer to map the output of the LSTM layer to binary classification decision. We use binary cross-entropy as the loss function, Adam optimizer and accuracy as the evaluation metric.
- The test accuracy obtained for RNN-LSTM model is 82.14 percent. The results can be visualized using the following heatmap.

D. mBERT

- * BERT stands for Bidirectional Encoder Representations from Transformers.
- * mBERT or multilingual BERT (BERT multilingual base model (cased)) is an open-source pre-trained transformer model trained in a self-supervised manner on a multilingual text corpus.
- * BERT was pre-trained for the purpose of Masked Language Modeling (MLM) and Next Sentence Prediction (NSP).
- * It uses Transformer attention mechanism which helps in comprehending ambiguous language in text, by learning the context.
- * We read the dataset and extract sentences into lists. Then we tokenized input sentences using BertTokenizer function.
- * We created a TensorDataset and split the dataset into training, validation and testing subsets in a ratio of 80:10:10.
- * We trained the model and used Adam optimizer for fine-tuning.
- * The test accuracy obtained from mBERT model is 82.74 percent. The results can be visualized using the following heatmap.



VIII. CONCLUSION AND FUTURE WORKS

We observe that the highest accuracy was obtained for the pre-trained Multilingual BERT(mBERT) and it outperformed the other models. It gave an accuracy of 93.75 percent. However, the accuracies of the other models can be improved by training them on a larger data-set. As a part of our future works, we are planning to explore the detection of other emotions like Sarcasm and Irony in Kannada code mixed and Kannada transliterated text.

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